DAVID JOHN NIEDZWIECKI JR.

Principal AI Systems Architect, Backend Developer, & Industrial Systems Engineer

Vanceboro / New Bern, NC | 252-259-0884 ofthefirstlight@gmail.com | **Github:** github.com/BiggerRoarThnU/Geminiology **OSF Academic Portfolio:** osf.io/s7f2d | **Enterprise:** SovereignNexus LLC | **UEI:** K5DALREZFGH6 | **CAGE:** 1AQG5

PROFESSIONAL SUMMARY

A systems architect, backend developer, and certified industrial fabricator specializing in deterministic edge AI deployments, secure local APIs, and low-latency system automation. Founder of SovereignNexus LLC, engineering air-gapped agentic environments operating within strict hardware limits (8GB/16GB RAM) and enforced by hardware-level safety thresholds. Combines production edge computing with military-grade structural fabrication experience (MCB Camp Lejeune). Built upon the principle of deterministic execution—treating code and data pipelines with the same tolerance standards, stress limits, and verification required in heavy structural engineering. Fully vetted and registered for federal and commercial contracting.

CORE TECHNICAL COMPETENCIES

- **Edge AI & Cognitive Systems:** Local LLM Deployment (Ollama, vLLM, Gemma, Llama), Multi-Agent MoE Routing, RAG, 1.58-Bit Ternary Quantization, Vector Mapping, Prompt Security & Sanitation.
- **Backend Systems Engineering:** Python (FastAPI, Flask), JSON-RPC 2.0, RESTful APIs, SQLite (WAL Concurrency), NDJSON streaming, Linux Administration (Debian/Lubuntu), Bash Automation, GitLab CI/CD.
- **Hardware Telemetry & Security:** Zero-Copy Virtual Memory Mapping (mmap), Cryptographic Auditing (SHA-256 fixity tracking), ECDSA validation, System Telemetry Daemons (RAM/CPU/Thermal Governor).
- **Industrial Fabrication & Logistics:** Structural Welding (SMAW, GMAW, FCAW, GTAW), Plasma Arc & Oxy-Fuel cutting, Blueprint Reading & Layout Spacing, AWS/ASME/API Standards, Warehouse Logistics.

KEY TECHNOLOGICAL ACHIEVEMENTS

- **Deterministic Edge Agent Environments:** Built zero-leak, memory-mapped (mmap) virtual paging buffers executing multi-agent pipelines offline on 8GB/16GB RAM commodity hardware; validated with 104 passing local test suites.
- **12-Node Mixture of Experts (MoE) Swarm Router:** Engineered a lightweight task-distribution shell quantizing inbound inputs to a 1.58-bit median scale, dispatching across 12 background micro-agents while constraining active heap allocation to <18.7 MB. • **Hardware Thermodynamic Governor:** Developed a hardware-level safety monitor that continuously reads system heat flux; initiates a controlled power shutdown above 105°C to prevent hardware faults, bit flips, and state drift.
- **Computer Vision Media Verification Pipeline:** Authored nexus_artifact_enhancer.py using OpenCV to ingest physical artifacts, normalize backgrounds, evaluate Laplacian variance against synthetic degradation, and record cryptographic SHA-256 hashes locally. • **Cryptographic Checkpoint Ledger:** Deployed a local Write-Ahead Logging (WAL) SQLite system (nexus_checkpoints.db) that binds sequential agent states using SHA-256 hashes and ECDSA signatures, ensuring an immutable local audit trail. • **Security Disclosure (HackerOne):** Identified and reported an active session leak exposing authentication tokens within high-frequency local logs (Report #3618959).
- **Published Technical Specifications:** Authored four published architectural papers cataloged on Zenodo and the Open Science Framework (OSF) covering deterministic runtime bounds and agentic safety controls.

PROFESSIONAL EXPERIENCE

SovereignNexus LLC | New Bern, NC — Principal AI Systems Architect & Founder (March 2026 – Present) •

Architected the Geminiology edge framework, deploying multi-agent models locally to eliminate cloud dependency and protect proprietary data integrity.

- Built sovereign_defense_core.py, a high-throughput FastAPI proxy sanitizing inputs via compiled regex expressions and verifying vector similarity thresholds.
- Designed asynchronous data chunking daemons (oracle_server.py, chunk_ingester.py) to parse unstructured technical and compliance documents into indexed SQLite tables.
- Authored master_t7_sync.sh utilizing rsync to mirror and hash active workspace code and system logs directly to encrypted external NVMe drives.
- Completed registration for SovereignNexus LLC on SAM.gov (UEI: K5DALREZFGH6 | CAGE: 1AQG5) to pursue federal contracting opportunities.

Greenewood Machine and Fabricators | Pollocksville, NC — Structural Welder & Fabricator (Federal Defense Contractor) (2019 – 2020)

- Obtained DBIDS clearance to execute structural fabrication and installation on-site at MCB Camp Lejeune and MCAS New River (MOUT Urban Training Facility).
- Operated scissor lifts, lulls, and overhead rigging to remove and set heavy steel structures up to three stories high in compliance with OSHA regulations.
- Performed structural joins and cuts using SMAW, GMAW, oxy-fuel, and plasma systems under strict AWS and ASME inspection codes.
- Interpreted complex mechanical blueprints to calculate material cut lists, layout parameters, and load-bearing requirements.

Binall Construction SDN BHD | Hubert, NC — Systems Laborer & Carpenter (2018 – 2019) • Read architectural framing plans and verified precision spatial measurements to reduce material waste on active job sites. • Coordinated structural removals, load-bearing timber framing, subflooring, and finished wallboard installations.

Labor Finders | Multi-State — Industrial Carpenter & Site Specialist (2017 – 2018)

- Interpreted technical schematics to build temporary forms, framing, and site foundations across commercial construction projects; managed material staging logistics.

Staples Warehouse | Brandon, FL / Plainfield, CT — Logistics & Heavy Equipment Operator (2015) • Operated industrial forklifts and order pickers to navigate high-density layouts safely; logged inbound freight manifests via terminal systems.

ESPN Sports Center (Via Martin Laivero) | Bristol, CT — Heavy Equipment Operator (2010 – 2014) • Operated Bobcat loaders and clearing machinery under severe weather conditions to keep critical broadcast facilities and transport arteries operational.

EDUCATION & CREDENTIALS

- **B.S. in Computer Science (In Progress)** | Western Governors University (WGU) — Focus: Discrete Mathematics, Data Structures & Algorithms, OS Architecture.
- **AI Systems & Security Specializations (In Progress)** | Google AI Professional Credentials & Coursera — Focus: Local model deployment, input defense, API architecture.
- **Industrial Welding & Thermal Fabrication Certificate** | Career Source Tampa Bay (2017) — Focus: Multi-process structural welding (SMAW/GMAW/FCAW/GTAW), AWS/ASME standards.

2017 Career Source Tampa Bay Career Prep Center –Tampa, FL

Welding and Fabrication Training

- **Non certification program** covering Thermal Cutting (oxy-Fuel, Plasma), Shielded Metal Arc Welding (SMAW), Gas Metal Arc Welding (GMAW), Flux Cored Metal Arc Welding (FCAW) and Gas Tungsten Arc Welding (GTAW) covering the entry level understanding of the process fundamentals. FCAW utilized both gas shielded and selfshielded filler materials. Both knowledge based and performance based testing were utilized to grade student progress.
- **Program entailed OSHA 10 training**, welding safety, equipment setup, and tear down, and hands-on performance with surface, fillet, and lap welds in all four welding positions on flat plate and tee joints for SMAW, GMAW , FCAW, and GTAW.
- **Also included introductory blueprint reading**, construction math, measuring and inspection techniques, and interpretation of welding symbols and electrode classifications as utilized in the fabrication industry.
- **Introductory welding** procedure and welding performance requirements as related to the AWS, ASME, and API fabrication codes.